MANAGEMENT SCIENCE
Vol. 36, No. 2, February 1990
Printed in U.S.A.

OPTIMAL BUNDLE PRICING*

WARD HANSON AND R. KIPP MARTIN
Krannert Graduate School of Management, Purdue University, West Lafayette, Indiana 47906
Graduate School of Business, 1101 *East* 58*th Street, University of Chicago, Chicago, Illinois* 60637

Bundle pricing is a widespread phenomenon. However, despite its importance as a pricing tool, surprisingly little is known about how to find optimal bundle prices. Most discussions in the literature are restricted to only two components, and even in this case no algorithm is given for setting prices. Here we show that the single firm bundle pricing problem is naturally viewed as a disjunctive program which is formulated as a mixed integer linear program. Multiple components, and a variety of cost and reservation price conditions are investigated with this approach. Several new economic insights on the role and effectiveness of bundling are presented. An added benefit of the solution to the bundle pricing model is the selection of products which compose the firm's product line. Computational testing is done on problems with up to 21 components (over one million potential product bundles), and data collection issues are addressed.
(MARKETING—PRICING, PRODUCT POLICY; MATHEMATICAL PROGRAMMING—MIXED INTEGER LINEAR)

1. Introduction

Pricing a product line is a difficult task. This is especially true when the extent of the product line must be determined simultaneously with the prices to be charged. Such a situation arises naturally when the products or services being offered are comprised of components. For example, a software company faces the problem of pricing a product line composed of multiple software modules. Ranging from statistics to graphics to data base management, each module provides a unique set of features. Different customer segments attach different values to each of these components. Some customers will want to purchase the full product line for complicated tasks, while other customer types are only interested in a specific component. Costs to the firm may also vary widely between components. Internally produced software will have very low incremental production costs, while components licensed from other software producers can have high incremental costs due to the royalty rates the firm must pay. The software company must decide on a pricing strategy for the individual components and for bundles of components.

In arriving at this pricing strategy, the firm must properly balance multiple incentives. Should prices encourage customers to select a wide range of offerings, including components which they do not value highly, or should prices encourage segmented purchasing at increased prices? Can the firm channel additional orders into those components which it produces internally, without eliminating customers who highly value the externally produced components? Should all the components be offered, or will this lead to excessive cannibalization within the product line and high production costs?

These are questions of *bundle pricing*. In this paper we investigate how a single firm, facing segmented customer demand and product specific costs, can determine optimal product line breadth and pricing. The methodology we develop can analyze a considerable range of bundling issues. It also shows substantial promise for a wider class of problems than just bundling, namely firm optimization when explicit attention is paid to customers' maximization of net benefits. These "self-selection" problems occur in such diverse areas as sales force compensation schemes and the dynamic pricing of durable goods.

* Accepted by Jehoshua Eliashberg; received June 17, 1987. This paper has been with the authors 7 months for 3 revisions.

0025-1909/90/3602/0155$01.25

The major concern of previous studies of bundling has been demonstrating the demand side incentives which cause bundling to be profitable. These include negative correlations in customer segment demands, complementarity in consumption, and uncertainty in the valuations of the quality of the goods. Bundling can also be profitable due to cost conditions. We briefly review each of these incentives.

Stigler (1963) is the most influential early study of bundling. In his framework, upon which most studies have been built, demand is represented as explicit customer segments. In the Stigler model, the relevant customer demand information is captured by a vector of reservation prices for the products. A customer will choose the product which maximizes individual surplus, which is defined as the difference between the reservation price and the product price. Both reservation prices and firm production costs are assumed to be additive.

Stigler used this setting to demonstrate that a motion picture distributor could raise profits by refusing to lease individual movies, and would prefer to only lease movies in multiple movie packages. What makes bundling profitable is a pattern of negative correlation in the reservation prices for individual movies, while the joint evaluation is similar.

Adams and Yellen (1976) utilize the same framework as Stigler to demonstrate that a firm would typically want to offer both the bundle and the separately priced components, a situation they term "mixed bundling." Their major change is to consider more than two customer segments. By increasing the number of customer segments, they demonstrate cases where it is desirable to serve highly asymmetric demanders with individual components while targeting the bundle toward the less extreme "middle" segments. This dominates pure bundling, as the extreme demanders have a low bundle valuation due to their indifference toward the second component.

Schmalensee (1984) modifies the basic Stigler framework by representing customer reservation prices as a bivariate normal distribution in the positive quadrant. This produces a continuum of customer segments. He retains the two component assumption, and bundle reservation prices are again additive. Two major results of his approach are a confirmation of the profitability of bundling when there is negative correlation, and the benefits of mixed bundling over a restriction to pure components or pure bundling.

None of these papers show how to calculate optimal bundle prices. In both Stigler and Adams and Yellen, the cases are sufficiently simple that optimal prices can be obtained by inspection. In Schmalensee, the special properties of the customer distribution allow a simulation approach.

A different framework and rationale for bundling is given by Telser (1979). Telser stresses the importance of complementarity between products as a source for bundling. Complementarity between components yields a valuation for the bundle which is super-additive, and this clearly enhances the chances that bundling is profitable. Our model also allows nonadditive reservation prices.

Uncertainty with regard to product quality can also be a basis for profitable bundling. Kenney and Klein (1983) argue that exact classification of the quality of some products, such as rough diamonds, may require excessive amounts of effort. A firm may find it more efficient to bundle these products together into a fixed package which must be accepted or rejected as an all-or-nothing proposition. Errors in classification of the individual stones will tend to cancel, leading to fewer rejections by customers of the bundles (called "sites"). We do not consider uncertainty in this paper.

The second broad category of incentives for bundling involves costs. Very little formal work has been done which relates cost assumptions to the desirability or form of bundling. Several qualitative observations on the benefits of bundling for controlling manufacturing setup costs, combined with industry examples, are contained in Porter (1985). These are not extended to a formal analysis. While not explicitly considering bundling, Baumol, Panzer and Willig (1980) have devoted extensive efforts to the study of economies of scope. As will be seen in §3.2, economies of scope can be a basis for profitable bundling.

In the next section we provide a practical method for calculating optimal bundle prices. The basis of the approach is to formulate the model as a mixed integer linear problem using disjunctive programming. The theoretical rationale for this approach is given along with computational results for a set of test problems based on actual survey data. An added benefit of the bundle pricing model solution is the selection of products to include in the firm's product line. Implications for managers based on model experiments are given in §3. Of prime concern are the relative benefits of reducing component cost versus raising customers' valuations for bundles. Reductions in production costs are always beneficial to a firm, but if economies of scope are present bundling may be necessary in order for the firm to operate profitably. Increasing customer valuations is more complicated, as we show how a costless increase in a customer's reservation price can lead to a fall in firm profits. In §4, we consider one of the most serious problems facing a product line manager addressing the bundling issue: the exponential growth in possible products which results from increasing the number of components considered. An algorithm for finding optimal solutions, subject to some additional assumptions beyond the model in §2, is given along with computational results. We conclude in §5 with possible model extensions.

2. Model Formulation

2.1. *Model Development*

In this section we formulate an optimization model for the bundle pricing problem. The model is based on the following sequence of events. A profit maximizing monopolist firm establishes a price list for all possible bundles of components. Customers observe the price list and make purchases which maximize their consumer surplus (which is equivalent to maximizing utility for the utility function discussed in §2.2). The model developed is from the firm's perspective, and the objective function represents the firm's profit. The behavior of the customers as maximizers of their surplus is treated in the model as constraints on the firm's objective function.

Problem Parameters.

$S = \{1, \ldots, n\}$ is the index set of component items.

h = index of components, $h = 1, \ldots, n$.

i = index of component bundles, $i = 1, \ldots, L = 2^n - 1$.

k = index of customer segments, $k = 1, \ldots, M$.

R_{ki} = reservation price of customer segment k for bundle i, $R_{k\{h\}}$ denotes the reservation price of the bundle which consists of only component h.

c_{ki} = cost of supplying one customer of segment k with bundle i.

N_k = number of customers in segment k.

$B(i)$ = the set of components which define bundle i.

$I \subseteq \{1, \ldots, L\}$ is an index set of bundles.

Decision Variables.

p_i = price assigned to bundle i, $\mathbf{p} = (p_1, \ldots, p_L)$.

θ_{ki} = 1 if customer segment k selects bundle i, 0 otherwise.

s_k = consumer surplus obtained by customer segment k.

Auxiliary Variables.

s_{ki} = the consumer surplus of a customer in segment k if that customer selects bundle i.

z_{ki} = the marginal revenue generated from a customer in segment k if that customer selects bundle i.

p_{ki} = the price a customer in segment k pays for selecting bundle i.

Our model is a generalization of the approach of Stigler, as well as Adams and Yellen. The core assumptions are given below.

Model Assumptions.

1. The benefit of a duplicate component is zero, and component resale is not possible.
2. A single profit maximizing firm determines the price of every bundle.
3. Given these prices, every customer maximizes his/her consumer surplus. Consumer surplus equals reservation price minus product price. Purchasing nothing yields zero consumer surplus.
4. All customer segments face the same prices.
5. The reservation prices of all customer segments are known for all bundles.
6. There is *free disposal* of unwanted components. Thus $B(i) \subseteq B(j) \subseteq S$ implies that the reservation price of bundle j is greater than or equal to the reservation price of i.
7. Marginal costs of a bundle are *subadditive*. This means that given any bundle $B(i)$ and index set I, $B(i) \subseteq \bigcup_{j \in I} B(j)$ implies that the marginal cost of $B(i)$ cannot exceed the sum of the marginal costs for bundles $B(j), j \in I$.
8. Customers have zero *assembly transaction costs* for creating bundles from separately offered subbundles.

Unlike Stigler or Adams and Yellen, we make no restrictions on the number of components offered, the structure of reservation prices (i.e., strictly additive), or customer segment sizes.

Free disposal of unwanted components, assumption 6, is a restriction on allowable reservation prices which implies that no customer segment would strictly prefer *not* to have a component included for free. While not holding for all products or components (for example, power windows are disliked by some automobile buyers), it is important for the algorithm we present.

The same is true for the restrictions on cost contained in assumptions 7 and 8. Subadditive bundle costs are a form of (weak) economies of scope, where the presence of one component in a bundle does not raise the cost of including other components, and may make it cheaper. Zero assembly transaction costs for creating bundles, assumption 8, is an insistence that bundles are well specified. If costly assembly must be performed on components A and B to achieve bundle AB, then "AB" really contains the additional component C representing assembly. However, the firm may well find it profitable to only offer the items A, B, and ABC in its product line.

By assumption 3, each customer maximizes his or her surplus. Thus, a customer will buy every bundle which yields a positive surplus. However, the union of all the components in each bundle purchased also constitutes a bundle. For example, if there are two components, A and B, and each component has positive surplus, the consumer would purchase both A and B. For model formulation purposes, it is much easier to assume that the customer buys only one bundle. Thus we want to show that the customer is never better off buying more than one bundle. In order for this to be true prices must be subadditive. Indeed, if the price of AB *were greater* than the price of A plus the price of B the customer would prefer to buy A and B, over buying AB (since there are zero assembly costs by assumption 8). We first show that given assumptions 1–8, a firm can always maximize profits by using subadditive pricing. This result is then used in the proof of Proposition 1, which shows that we may assume that the customers purchase at most one bundle. Proofs appear in the Appendix.

LEMMA 1. *Given assumptions* 1 *through* 8, *there exists a profit maximizing set of prices which are subadditive.*

PROPOSITION 1. *Given assumptions* 1 *through* 8, *every customer will purchase exactly one bundle, or will not make a purchase.*

The proof of Proposition 1 is based on the fact that *every* bundle is priced and offered to all customers. Obviously it is impractical to package or list prices for $2^n - 1$ bundles when n is large. This does not cause an implementation problem. In the bundle pricing model given below, if there are M customer segments then at most M distinct bundles are part of an optimal solution. Only the optimal bundles need be offered as long as the optimization model considers every bundle in determining which M bundles to offer. Note the crucial distinction between the terms *offer* and *consider*. A bundle is considered in the model if that bundle can be part of an optimal solution. A bundle is offered when it is part of an optimal solution. Therefore, when offered the M bundles from the optimal model solution the consumers behave exactly as they would if given the complete list of optimal prices. Indeed, a key result of the bundle pricing solution is the choice of which products to offer out of the vast number of possibilities.

Given assumptions 1 through 8 and Proposition 1, the constrained optimization model for maximizing the firm's profit is given in (1) to (11). We refer to this as the optimal bundle pricing model.

Objective function

$$\max \sum_{k=1}^{M} \sum_{i=1}^{L} N_k z_{ki}, \tag{1}$$

Consumer surplus

$$s_k \geq R_{ki} - p_i, \qquad i = 1, \ldots, L; \qquad k = 1, \ldots, M, \tag{2}$$

Price subadditivity

$$p_i \leq \sum_{j \in I} p_j, \qquad \text{s.t.} \bigcup_{j \in I} B(j) = B(i), \qquad i = 1, \ldots, L, \tag{3}$$

Single price schedule

$$p_{ki} \geq p_i - (\max_{\substack{k=1,\ldots,M \\ i=1,\ldots,L}} \{R_{ki}\})(1 - \theta_{ki}), \qquad i = 1, \ldots, L; \qquad k = 1, \ldots, M, \tag{4}$$

$$p_{ki} \leq p_i, \qquad i = 1, \ldots, L; \qquad k = 1, \ldots, M, \tag{5}$$

Tightening constraints

$$s_k \geq \sum_{i=1}^{L} (R_{ki}\theta_{ji} - p_{ji}), \qquad k = 1, \ldots, M; \qquad j = 1, \ldots, M, \tag{6}$$

Single purchase

$$z_{ki} = p_{ki} - c_{ki}\theta_{ki}, \qquad i = 1, \ldots, L; \qquad k = 1, \ldots, M, \tag{7}$$

$$s_{ki} = R_{ki}\theta_{ki} - p_{ki}, \qquad i = 1, \ldots, L; \qquad k = 1, \ldots, M, \tag{8}$$

$$s_k = \sum_{i=0}^{L} s_{ki}, \qquad k = 1, \ldots, M, \tag{9}$$

$$\sum_{i=0}^{L} \theta_{ki} = 1, \qquad k = 1, \ldots, M, \tag{10}$$

$$p_i, p_{ki}, s_{ki} \geq 0, \qquad s_{k0} = 0, \qquad \theta_{ki} \in \{0, 1\}. \tag{11}$$

Constraint set (2) enforces the assumption that each customer maximizes surplus, s_k. This is done by requiring that the consumer surplus in the optimal solution be greater than or equal to the consumer surplus that could be achieved through the purchase of any other bundle at the optimal prices.

Constraints (3) are the subadditivity constraints. In (3) we write

$$B(i) = \bigcup_{j \in I} B(j), \tag{12}$$

instead of:

$$B(i) \subseteq \bigcup_{j \in I} B(j). \tag{13}$$

This is valid by Lemma 2, which shows the constraints in (13) with strict inclusion are redundant. Using strict set equality greatly reduces the number of constraints required in (3).

LEMMA 2. *If $B(i) \subset \bigcup_{j \in I} B(j)$ then there exists an optimal set of prices to* (1)–(11) *such that $p_i \leq \sum_{j \in I} p_j$.*

Because we prohibit explicit price discrimination between customer segments, if customer segment k buys bundle i and customer segment h buys the same bundle, then $p_{ki} = p_{hi}$. This condition is enforced by constraints (4) and (5). When $\theta_{ki} = 0$, then $p_{ki} = 0$ satisfies (4) and (5) since p_i need never exceed the maximum reservation price (over all customer groups and bundles). When $\theta_{ki} = 1$, then $p_{ki} = p_i$ and every customer segment buying bundle i pays price p_i.

Constraints (6) are redundant to the underlying mixed integer programming problem but not the linear programming relaxation. Because at most one θ_{ji} is positive (in the integer programming solution) for customer j, constraint (6) says that the surplus customer k receives for bundle i must be greater than or equal to the surplus possible from a bundle purchased by someone else. This tightens the linear relaxation of the problem because the sum on the right-hand side of (6) is over *all* bundles, whereas the right-hand side in (2) includes only one bundle. Constraint (6) cannot replace constraint set (2) since it is enforcing consumer surplus maximization for only those bundles actually purchased and does not include bundles which could be offered.

By Proposition 1 every customer will purchase exactly one bundle, or will not make a purchase. This is a disjunctive constraint because it says that a customer will buy nothing, or buy bundle one, or buy bundle two, . . . , or buy bundle L. The single purchase (or disjunctive) requirement is enforced in constraints (7)–(11), which involve the use of auxiliary variables. Disjunctive constraints can be modeled using mixed 0/1 linear programming. However, the well-known "Big M" (Wagner 1975, Hillier and Lieberman 1986, or Eppen and Gould 1985) method for modeling disjunctions as mixed 0/1 linear programs is often impractical for even small problems, because of the long computation times required for optimization. Recent papers by Balas (1985) and Jeroslow and Lowe (1984) propose an alternative methodology, which we utilize. Unlike the Big M approach, which requires only variables z_k, s_k and p_k, auxiliary variables z_{ki}, s_{ki} and p_{ki} are required for the disjunctive approach.

The key to understanding constraints (7)–(11) lies in constraint (10). This says that for each customer segment k exactly one θ_{ki} will equal one and the others zero. This implies that at most one p_{ki} and one s_{ki} are ever nonzero. Thus constraints (7)–(9) correctly define the revenue and consumer surplus that firm and customer k get when bundle i is purchased.

With the Big M method, constraints (7)–(9) are modeled as

$$z_k \leq p_i - c_{ki} + M(1 - \theta_{ki}), \tag{14}$$

$$s_k \leq R_{ki} - p_i + M(1 - \theta_{ki}). \tag{15}$$

(We use $\leq$ constraints instead of = since the maximization model will make z_k and s_k are large as possible.)

Compare (7)–(9) with (14)–(15) for a fractional θ_{ki}. Larger revenues and surpluses are feasible in (14)–(15) than (7)–(9). Thus, the linear programming relaxation is weaker in the Big M approach. The consequences of this are very significant for branch and bound algorithms. See Jeroslow and Lowe (1985).

For customer k, let W_k denote the set of feasible points defined by (7)–(11). Unlike the Big M formulations, the linear programming relaxation of (7)–(11) is equal to the convex hull of the mixed 0/1 points in W_k.

PROPOSITION 2. *Let $\bar{W}_k$ denote the polyhedron defined from* (7)–(11) *by replacing* $\theta_{ki} \in \{0, 1\}$ *with* $0 \leq \theta_{ki} \leq 1$. *Then for* $k = 1, \ldots, M$

$$\bar{W}_k = \mathrm{CONV}(W_k) \tag{16}$$

where CONV($*$) *denotes the convex hull of set* $*$.

PROOF. See Balas (1985) and Jeroslow and Lowe (1984).

If the objective function in (1) were optimized over only W_k, absolutely no enumeration would be required. Of course, other constraints such as (2) and (3) are present. Nonetheless, this modeling of the problem still has a tremendous effect on reducing the enumeration required, and thus greatly increases the size of problems which can be practically solved.

2.2. *Generating Model Inputs*

The bundle pricing model uses three different types of data as inputs. These are the reservation prices of all relevant customer segments for each of the bundles, the size of each of these segments of customers, and the cost of supplying a customer of a specific segment with a particular bundle. A substantial literature in the marketing, economics, statistical, psychological, and manufacturing literature has developed concerning the proper methodology for collecting such information. Here we discuss the specific data collection issues raised by bundling.

The traditional economic approach to customer choice among items in a product line, at specified prices, assumes that customers choose the item which maximizes their utility. Utility is a function of the attributes of the product chosen and the income remaining after purchase. In our model a customer maximizes the difference between bundle specific reservation and product prices. These approaches yield exactly the same predictions if utility functions are constrained to have the form given in equation (17), where the utility of customer k for bundle i at price p_i is linearly additive in income.

$$U_k(B(i), Y_k - p_i) = g_k(B_i) + \lambda_k(Y_k - p_i). \tag{17}$$

In this formulation, $g_k(B_i)$ represents the utility customer k would receive from receiving the bundle $B(i)$ at no cost. By free disposal it is nondecreasing in components, but may be nonlinear. The variable λ_k is the marginal utility of money, assumed positive, and Y_k is the level of pre-purchase cash. Following Kalish and Nelson (1988),

PROPOSITION 3. *If the utility for customer k is given by* (17) *then the bundle which maximizes* $R_{ki} - p_i$ *maximizes customer k utility, where*

$$R_{ki} = \frac{g_k(B(i))}{\lambda_k}. \tag{18}$$

Proposition 3 provides a consistent approach to determining reservation prices. Each bundle's reservation price is equal to the total utility contribution of the set of components divided by the marginal utility of money. Each component list of bundle i can be thought of as a vector of length n, with each element a binary variable indicating the presence or absence of that component.

Various market research techniques exist to elicit consumer valuations of products. One technique is to simply ask consumers for their reservation prices. A recent experimental study by Kalish and Nelson (1988) found consumers able to provide useful direct estimates of reservation prices. The reservation price approach was judged superior in explaining variance, but somewhat worse in predictive validity, as compared to more conventional conjoint measurement approaches such as Green and DeSarbo (1979), Goldberg, Green, and Wind (1984), and Green and Krieger (1985). For the conjoint measurement approach, equation (18) would be used to infer reservation prices.

The market research techniques cited above are based on individual responses. Although our model could in principle be solved on this individual level data, this is likely to be very inefficient and lead to unacceptable solution times. Thus, some form of aggregation from the individual responses is desirable. In the next section, cluster analysis is used for that purpose. As in any aggregation scheme, a balance must be struck between the consequences of loss of information and the benefits of increased convenience from larger segment sizes. For recent discussions of these points see Ogawa (1987) and Kalish and Nelson (1988).

Relevant costs for the model include costs which are segment specific and those which are common across segments for a particular bundle. Once segment membership has been determined, the segment specific cost determination is possible, especially for established product lines. These would include differential shipping expenses, support costs, etc. For the software example given in the introduction, extensive company records permit estimation of support and maintenance costs by type of user. In general, estimating the costs of the bundles, c_{ki}, is no different than estimating the marginal costs of any product line.

2.3. *Computational Results*

Thirty test problems were solved to demonstrate the tractability of the bundle pricing model formulation given in §2.1. The problems tested had four components and between five and ten customer segments. Recall that a four-component problem involves $2^4 - 1 = 15$ bundles to price. In §5 we present computational results on problems with 21 components (over one million products to price).

The data were generated from surveys taken in two MBA classes of 38 and 36 students, respectively. The survey was based on a questionnaire for a fictional company offering home services for urban professionals. The four service components were apartment cleaning, laundry and ironing, grocery shopping, and meal preparation. Any of these four components could be purchased individually, or in any bundle. The students were directly asked to state their reservation prices for each combination of services. Cluster analysis was used to group students into customer segments based on their reservation prices. A cluster analysis routine was run three times for each class. Students were grouped into five, eight, and 10 clusters (i.e. customer segments). The reservation prices used for each customer segment were the mean of the reservation prices of students within each cluster.

Both subadditive and additive marginal cost data were tested. Additive marginal costs are realistic if a different employee is used to perform each service, subadditive are realistic if the same employee provides several services. The marginal costs used were consistent with costs in the Chicago area for these types of services. In total, five sets of marginal cost data were used. All five sets of cost data were used for each class and each of the three cluster groupings, giving $(2)(3)(5) = 30$ test problems. Minimum, maximum, average times, and integrality gaps are reported for each set of five marginal costs.

The solution times in Table 1 are on a Digital Computer VAX 8650 running VMS

TABLE 1
Problem Size and Solution Times

		Variables		CPU Seconds			% Integrality Gap		
Problem	Rows	Cont.	0/1	Min	Max	Avg	Min	Max	Avg
$M = 5$	436	256	80	13.01	26.92	22.242	0	4.0	2.0
$M = 8$	706	400	128	51.81	222.09	110.49	0.8	6.8	3.2
$M = 10$	896	496	160	62.46	327.88	150.76	2.2	8.3	4.0
$M = 5$	436	256	80	14.4	24.17	20.45	1.6	2.3	2.0
$M = 8$	706	400	128	114.54	263.72	173.834	1.8	6.3	3.6
$M = 10$	896	496	160	452.73	838.62	544.464	1.7	5.7	3.6

using the LINGO modeling and mixed integer programming software (Cunningham and Schrage 1988). The times represent the total time required for matrix generation, solving the linear programming relaxation, finding the optimal integer solution, and proving optimality of that solution. The solution times indicate that problems with up to 10 customer groups and four components can be solved routinely on a machine of this power. This makes sensitivity analysis and scenario generation practical. A key reason for the tractability of our formulation is the quality of the linear programming relaxations. These values are given in the table in terms of integrality gap. The percent integrality gap is defined as

$$\left[\left(\frac{\text{Optimal LP Obj.} - \text{Optimal IP Obj.}}{\text{Optimal IP Obj.}}\right)(100)\right].$$

There are a number of important examples with the same orders of magnitude as the problems solved in Table 1. For example, in a vacation package the components might be a hotel, rental car, air fare, meals, and tours. In selling a personal computer relevant components might include a monitor, printer, hard disk, and the basic unit. When the number of components is too large for the model of (1)–(11), the technique developed in §4 is appropriate.

3. Bundling Applications

3.1. *Bundling and Reservation Prices*

Utilizing bundling forces a firm to think about its entire product line when determining prices. It must also be alert to the effects of its other marketing variables, as some of the most basic intuitions carried over from single product models need not hold in the more complicated bundling regime. Specifically, a costless increase in the reservation prices of all customer segments for a bundle can lower firm profits.

Table 2 contains the reservation prices for a 2-component-3-customer segment problem. Solving the model leads to component A not being offered, a price of 95 for component B, 195 for the bundle AB, and a profit of 6550. Consider an advertising campaign or slight product modification which makes component B more valuable as a standalone component. We assume that it does not affect the reservation prices for the other products (thus, there is a Pareto increase in reservation prices), and it is costlessly achieved. This alteration raises the reservation price of segment 1 for component B to 100, for segment

TABLE 2
Pareto Increase in Reservation Prices

Customer	Reservation Prices			Segment Size
	Component A	Component B	Bundle AB	
Segment 1	100	95	195	30
Segment 2	50	95	145	100
Segment 3	80	80	195	80
Marginal Cost	50	90	140	

2 to 96, and segment 3 to 85. All other reservation prices remain unchanged. Resolving the model leads to a fall in profits to 6210, with component A still omitted, component B priced at 96, and the price for the bundle AB equal to 191.

The intuition behind this result rests on the differential margin of the products, and the variation in segment sizes. Note that bundle AB has a relatively large margin compared to component B. This margin is threatened by the rising reservation price for component B, for if prices do not change segment 1 will switch to this less profitable product. To prevent this switch the firm must lower the prices of bundle AB. It is better to do this than to discontinue component B, as the size of segment 2 is substantial.

Although this result appears counterintuitive, it is consistent with some actions which firms take to avoid product cannibalization within their product line. The business press from time to time will note with exasperation the tendency of a firm to "cripple" a product by deliberately omitting key features. In our model this translates into the firm altering component B in order to reduce segment 1's valuation (relative to segments 1 and 2) of this component from the value of 100.

This phenomenon has interesting implications for the general marketing support given to a product line. The increase in reservation prices contained in Table 2 might occur as the result of an advertising and promotional campaign for component B. As the example shows, *even if the advertising and promotion was free* it would lower profits.

3.2. *Bundling and Costs*

Increasing customer reservation prices may reduce firm profits. This is not the case with costs. Lowering costs on any bundle can never reduce profits, and if the bundle is currently being sold, leads to increased profit. The key difference is that, unlike changes in reservation prices which affect consumer surplus constraints, changing costs has no effect on the feasible region. If a solution is feasible, then this solution remains feasible after a cost change. Although it is to be expected that increasing reservation prices will normally lead to higher profits, lowering costs has a more consistent effect on firm results than does raising reservation prices.

Not only does component cost reduction never hurt profitability, but bundling is an important method of controlling costs in a number of industries. Comprehensive products often have costs which are subadditive, that is, the more complete bundles have lower marginal costs. The importance of subadditive costs is shown below.

Perhaps the most important alternative pricing approach to a bundling strategy is separate pricing for each of the individual components, with customers "mixing and matching" for their most desired total product. Pure components pricing limits the firm to indirect pricing of the more comprehensive bundles, as bundle prices are constrained to equal the sum of the separate component prices (the pure components strategy is obtained by replacing the price subadditive inequalities in (3) with equalities). Adams and Yellen provide examples showing mixed bundling is preferable to pure components pricing. In their examples they assume marginal costs and reservation prices are additive. With these assumptions, if the firm can make positive profit using mixed bundling, then the firm can make positive profit from pure component pricing. We show if marginal production costs are subadditive, then bundling may be necessary in order to make positive profit (see Table 3).

In this example all marginal costs are additive, with the exception of bundles AB and ABC. There are economies of scope from producing components A and B together, and moderate additional savings from combining all three components. These are also the only bundles which can yield strictly positive profits (component B can just break even). All reservation prices are additive.

First consider the case when the segment sizes are all equal to some level N. Solving the bundle model results in the sale of bundles (B, AB, ABC) at prices (70, 125, 200) for a profit of $40N$. Component B just breaks even, and is purchased by segments 1 and 3. Customer segment 2 buys bundle AB, and customer segment 4 buys bundle ABC.

The optimal pure components price solution is identical, with the exception that the price for component B must be dropped to 60. This results in a loss of 10 on each sale of component B, and lowers profits to $20N$. This loss must be incurred, due to the lack of pricing flexibility. The firm cannot lower the price of component A below 65 without leading to switching by segments 2 and 4 out of the profitable bundle sales into losses on component A. But if component A is priced at 65, then component B must be set equal to 60 in order to sell AB at 125. If the firm chooses not to sell AB, it can avoid the loss on component B by setting a price for B equal to 70. However, it then must set a price of 60 for component C in order for bundle ABC to equal 200. This would cause both segments 3 and 4 to choose the (now) unprofitable component C. All of these problems follow from the price equality conditions imposed by pure components pricing.

The situation grows worse if we vary the relative segment sizes. In particular, let the customer distribution now equal ($2N$, N, $2N$, N). The bundle solution is unchanged by this growth in size of segments 1 and 3. Profits are still equal to $40N$, and the price and purchase patterns remain the same. The pure components solution is also the same in terms of prices and purchase patterns, but now firm profit equals 0. The losses on component B completely offset the gains from the other products. Indeed, were either segments 1 and 3 to grow more, the firm would be forced to shut down and make no sales (other than the trivial case of selling only component B at a price of 70).

This example dramatizes the role that bundle pricing can play in allowing a firm to benefit from subadditive costs. Without the flexibility of setting the bundle prices separately, the firm may be unable to encourage customers to purchase those products which are the cheapest to produce. It is blocked in this attempt by customers self-selecting into the more attractive, but more expensive to produce, individual components.

TABLE 3

Bundling and Costs

	Reservation Prices						
Customer	A	B	AB	C	AC	BC	ABC
Segment 1	40	70	110	20	60	90	130
Segment 2	65	60	125	30	95	90	155
Segment 3	40	70	110	75	115	145	185
Segment 4	65	60	125	75	140	135	200
M.C.	70	70	110	80	150	150	175

4. Pricing Large Numbers of Components

The number of constraints and variables in the bundle pricing model formulation (1)–(11) grows exponentially with the number of components. Therefore, an alternative to directly optimizing (1)–(11) is required when there are a large number of components. In this section we develop a solution procedure which grows linearly with the number of components.

This procedure requires that the reservation prices and marginal costs of all bundles be strictly additive. While this may seem restrictive, we show that by making appropriate restrictions on customer reservation prices an important class of problems with nonadditive reservation prices is easily converted into problems with additive reservation prices. The model of this section is then applied to the resulting problem. Moreover, the demand restrictions used for conversion are often the practical restrictions one must impose on large problems in order to determine the consumer data empirically. We next state the demand restrictions, show how to construct an equivalent model with additive reservation prices, and illustrate using an example with 21 components.

For the large n case there is often a *key component* which plays a unique role, and has a strong nonlinear influence on all other components. An example is a basic house, which is augmented by a large number of options. Other examples of key components are a basic car, which can be bundled with a range of additional features, and a basic computer, which can be enhanced with a number of components.

Central to the idea of a key component is the notion that peripherals and accessories have very little value in the absence of the key component. Thus, the inclusion or absence of the key component implies a significant nonlinearity in reservation prices for various bundles. However, this nonlinearity often has the property that an "adjusted" set of additive reservation prices can be found by defining the adjusted reservation prices of a component to be equal to the value it adds to the key component.

DEFINITION. Component n is a *key component* if and only if for every bundle i and customer segment k

$$R_{ki} = R_{k\{n\}} + \sum_{\substack{h \in B(i) \\ h \neq n}} R'_{k\{h\}} \qquad \text{when} \qquad n \in B(i) \tag{19}$$

where the $R'_{k\{h\}}$ are defined by

$$R'_{k\{h\}} = R_{k,\{h,n\}} - R_{k\{n\}}, \qquad h = 1, \ldots, n-1. \tag{20}$$

The reservation prices defined in (20) are called the adjusted reservation prices. The adjusted reservation prices are used to define a new model with reservation prices Q_{ki} where

$$Q_{k\{h\}} = R'_{k\{h\}}, \qquad h = 1, \ldots, n-1,$$

$$Q_{k\{n\}} = R_{k\{n\}},$$

$$Q_{ki} = \sum_{h \in B(i)} Q_{k\{h\}} \qquad \text{for cardinality} \qquad B(i) > 1.$$

By construction, the reservation prices Q_{ki} are additive and $Q_{ki} = R_{ki}$ if bundle i contains the key component. The following proposition gives sufficient conditions for the bundle pricing model with reservation prices Q_{ki} to yield the optimal set of prices for the bundle pricing model with reservation prices R_{ki}.

PROPOSITION 4. *An optimal set of prices to the bundle pricing model using reservation prices Q_{ki} is an optimal set of prices to the bundle pricing model with reservation prices R_{ki} if n is a key component and*

(i) $Q_{ki} \geq R_{ki}$ *for all bundles i such that $n \notin B(i)$,*

(ii) *for all customer segments k and bundles i with $n \notin B(i)$ we have $c_{ki} > R_{ki}$,*
(iii) *every bundle selected in the model using reservation prices Q_{ki} contains component n.*

When solving the large n case, we assume conditions (i)–(iii) of Proposition 4 and also (iv) that marginal costs are additive. Conditions (i) and (ii) of Proposition 4 are reasonable for many interesting cases, as the reservation prices for most bundles without the key component will be very small. For testing of the model in this section, we consider bundles representing various housing alternatives. An example of conditions (i) and (ii) would be a very low valuation for a formal dining room without the key component of the house. Given additive reservation prices, we can optimize the bundle pricing model for large n. In the following discussion, we assume that the original reservation prices R_{ki} are strictly additive, or that the R_{ki} are adjusted reservation prices which are strictly additive. Condition (iii) is enforced by only offering bundles which contain the key component.

An iterative approach is used to solve the bundling problem with large n. This iterative approach involves two simple ideas. First, since it is not practical to create a variable for every possible bundle, we explicitly list a very small subset of the potential bundles. Second, we use the assumptions of additive reservation prices and marginal costs and allow each customer to construct an alternative preferred bundle using the n individual components if the explicitly listed bundles are not appealing (the reservation prices and marginal costs of alternative bundles are simply the sum of the individual component reservation prices and costs, respectively).

Since any of the $2^n - 1$ bundles can be constructed from the n individual components, the mathematical model does implicitly consider every bundle. We refer to these alternative bundles as *composite bundles*. Composite bundles are used for *modeling purposes only*—the customer does not physically construct a composite bundle by purchasing multiple components. If the solution to the model has a customer constructing a composite bundle, then the firm offers this bundle for sale and the customer still purchases exactly one bundle. The following additional notation is used in the new model.

Additional Variables and Parameters.
$\alpha_{kh} = 1$ if customer k selects component h to be in the composite bundle, 0 otherwise.
p_{kc} = price consumer k pays for a composite bundle.
$E(L) \subseteq \{1, \ldots, L\}$ is the subset of bundles which are considered explicitly.
$r_{hi} = 1$ if bundle $i \in E(L)$ contains component h, 0 otherwise; this is a parameter.
z_{kc} = marginal revenue generated from a customer in segment k if that customer purchases a composite bundle.
s_{kc} = consumer surplus generated from a customer in segment k if that customer purchases a composite bundle.

The optimization model used in the algorithm to solve the bundling problem for large n is given below. We refer to this formulation as the *relaxed bundle pricing model*. The term relaxed is explained later.

Objective function

$$\max \sum_{k=1}^{M} N_k\Big(z_{kc} + \sum_{i \in E(L)} z_{ki}\Big), \tag{21}$$

Consumer surplus

$$s_k \geq R_{ki} - p_i, \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{22}$$

$$s_k \geq \sum_{h=1}^{n} R_{k\{h\}}\alpha_{lh} - p_{lc}, \qquad k = 1, \ldots, M; \qquad l = 1, \ldots, M, \tag{23}$$

为何区分k&l?

Single price schedule

$$p_{ki} \geq p_i - (\max_{\substack{k=1,\ldots,M \\ i=1,\ldots,L}} \{R_{ki}\})(1-\theta_{ki}), \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{24}$$

$$p_{ki} \leq p_i, \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{25}$$

Tightening constraints Pkc不需要设置上下界吗?

$$s_k \geq \sum_{i=1}^{L} (R_{ki}\theta_{ji} - p_{ji}), \qquad k = 1, \ldots, M; \qquad j \in E(L), \tag{26}$$

Single purchase

$$z_{kc} = p_{kc} - \sum_{h=1}^{n} c_{k\{h\}}\alpha_{kh}, \qquad k = 1, \ldots, M, \tag{27}$$

$$z_{ki} = p_{ki} - c_{ki}\theta_{ki}, \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{28}$$

$$s_{kc} = \sum_{h=1}^{n} R_{k\{h\}}\alpha_{kh} - p_{kc}, \qquad k = 1, \ldots, M, \tag{29}$$

$$s_{ki} = R_{ki}\theta_{ki} - p_{ki}, \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{30}$$

$$s_k = s_{kc} + \sum_{i \in E(L)} s_{ki}, \qquad k = 1, \ldots, M, \tag{31}$$

$$\theta_{kc} + \sum_{i \in E(L)} \theta_{ki} = 1, \qquad k = 1, \ldots, M, \tag{32}$$

$$\alpha_{kh} \leq \theta_{kc}, \qquad k = 1, \ldots, M; \qquad h = 1, \ldots, n, \tag{33}$$

这里composite bundle一定包含component h的目的

$$\alpha_{kh} \leq \alpha_{kn}, \qquad k = 1, \ldots, M; \qquad h = 1, \ldots, n-1, \tag{34}$$

Prevent cycling

$$\sum_{h=1}^{n} r_{hi}\alpha_{kh} - \sum_{h=1}^{n} (1 - r_{hi})\alpha_{kh} \leq (\sum_{h=1}^{n} r_{hi}) - 1, \qquad i \in E(L); \qquad k = 1, \ldots, M, \tag{35}$$

$$s_{kc}, p_{kc}, s_{ki}, p_{kc} \geq 0, \qquad s_{k0} = 0, \qquad \alpha_{kh} \in \{0, 1\}, \qquad \theta_{ki} \in \{0, 1\}. \tag{36}$$

Compare each set of constraints in the relaxed bundle pricing model with the corresponding set of constraints in the bundle pricing model. The consumer surplus constraints given by (2) are for every possible bundle. In constraints (22)–(23) there are surplus constraints for only the explicitly listed bundles in set $E(L)$ and the composite bundle. The subadditivity constraints (3) are dropped from the relaxed model because they require knowing which components every bundle contains. The components in the composite bundle are not known before the model is solved. The single price schedule constraints given by (4)–(5) are enforced in (24)–(25) for only the explicitly listed bundles in set $E(L)$. Similarly for the tightening constraints.

The single purchase constraints for the relaxed bundle pricing model are similar to those of the bundle pricing model, except constraints defining the composite bundle have been added. Constraint (27) defines the revenue for the composite bundle. Constraint (29) defines the consumer surplus associated with the composite bundle. Constraints (28) and (30) are identical to (7)–(8), respectively, with the exception that they are defined only over $E(L)$. Constraint (31) is analogous to (9) but allows for slack from the composite bundle. Constraint (32) enforces the disjunctive condition that customer segment k purchases exactly one of the explicit bundles, purchases a composite bundle, or purchases nothing (the null bundle is considered as an explicit bundle). Constraint (33) is a forcing constraint. If any individual component is selected by customer k, then the composite bundle is the bundle selected by customer k. Constraint (34) enforces condition (iii) of Proposition 4, and requires the key component to be present in every bundle (assume component n to be the key component). Finally, constraint (35) enforces the condition that the composite bundle is not identical to any of the explicitly listed bundles. This constraint is based on the cut of Balas and Jeroslow (1972).

PROPOSITION 5. *An optimal solution to the relaxed bundle pricing model*

1. *yields an optimal solution value which is an upper bound on the optimal solution value to the bundle pricing model, and*
2. *is an optimal solution to the bundle pricing model if the solution* (i) *satisfies price subadditivity*, (ii) *maximizes consumer surplus for the given prices*, *and* (iii) *no two consumers pay different prices for the same composite bundle*.

The model given by equations (21) through (36) is called a relaxed model because feasible solutions to this model may not be feasible for the bundling model with all bundles listed explicitly. The intuition behind Proposition 5 is that if we find a solution to a model with some of the constraints dropped then we have an upper bound (assuming maximization) on the original problem. If we are fortunate, and the solution to the relaxed problem satisfies the constraints that were dropped, then we have an optimal solution to the original problem.

After optimizing the relaxed bundle pricing model one has an upper bound on the optimal solution value of the bundle pricing model. In order to see if this solution is also feasible to the full bundle pricing model the conditions of part 2 in Proposition 5 must be checked. Condition (iii) is trivial to check by inspection while conditions (i) and (ii) require a more systematic method for realistic sized problems. The algorithms for accomplishing each of these checks are given in the Appendix.

The relaxed bundle pricing model and the subproblems for testing for subadditivity and maximum surplus suggest the following algorithm for solving the bundle pricing problem for large n.

Optimal Bundle Pricing Algorithm.

Step 1. Solve the Relaxed Bundle Pricing Model given by (21)–(36). The optimal solution value is an upper bound on the optimal solution value to the bundle pricing problem. Go to Step 2.

Step 2. Solve the Price Subadditivity Subproblem (see Appendix). If the prices in the solution to (21)–(36) are subadditive go to Step 3. Otherwise, (i) add the appropriate subadditivity constraints as indicated by the subproblem, and (ii) add any composite bundles j for which $\delta_j = 1$ or $\gamma_{jl} = 1$ to the list of explicit bundles and return to Step 1.

Step 3. Solve the Consumer Surplus Maximization Subproblem (see Appendix). If the solution to (21)–(36) maximizes consumer surplus for given prices, then go to Step 4. If not, add as explicit bundles those bundles defined by $\pi_{ki} = 1$ and return to Step 1.

Step 4. Check for Price Discrimination Among Composite Bundles. Two customer segments may create the same composite bundle. If so, they must pay the same price for the composite bundle by model assumption 4 given in §3. If two or more customer segments create the same composite bundle but pay different prices, add these composite bundles as explicit bundles and return to Step 1. Otherwise stop, the current solution is optimal.

The model converges after a finite number of steps since at least one explicit bundle is added at each step, by constraint (35) the same explicit bundle can be added only once (i.e. there is no cycling), and there are a finite number of bundles.

This could lead to a very large number of bundles. In order to test the convergence of the algorithm 40 test problems were generated. The test problems had 20 components in addition to the key component and four customer segments. For each run a different random number seed is used to generate uniform (0, 1) random numbers. The random number generator used is the one described in Bratley, Fox, and Schrage (1983). Each customer group requires 22 random numbers. The first random number is used to select the number of customers in the group. The number of customers per segment is assumed to be uniformly distributed between 1 and 10. The second random number is used in selecting a problem parameter α. The parameter α, used in reservation price generation, is in the open interval (0.8, 1.2). The remaining random numbers are used to generate reservation prices for the components. Components are considered to be cheap, moderate or expensive. The costs of these components are \$80, \$450, and \$800, respectively.

Customer reservation prices for these components are assumed to be uniformly distributed in the interval (\$100, \$500) $\times\ \alpha$, (\$500, \$1000) $\times\ \alpha$ and (\$1000, \$3000) $\times\ \alpha$, for cheap, moderate and expensive components, respectively. Reservation prices selected in this manner reflect, for each customer segment, correlation of reservation prices among cheap, moderate and expensive components. Some customers will tend to have high reservation prices on components while others will have low reservation prices.

In the first 20 runs we assumed that there were 10 cheap components, five moderate components and five expensive components. The cost of the key component was assumed to be \$65,000. Each customer group was given a reservation price of \$80,000 for the key component. In runs 21 through 40 the reservation price of the key component was given by \$80,000 $\times\ \alpha$. In runs 36 through 40 the cost of the key component was \$70,000. In runs 21 through 25 all components were cheap. In runs 26 through 30 all components were moderate and in runs 31 through 35 all components were expensive. Every run was made with two explicitly defined bundles. These were the null bundle and the bundle with every component present.

In all 40 runs the optimal solution was found and proved in *one iteration* of the optimal bundle pricing algorithm. We speculate that the large variety of potential bundles is actually contributing to the success of the algorithm. With so many possible bundles (2^{20}) available to the customer it is very unlikely that any two customer segments select exactly the same bundle. Hence, in Step 4 there is an extremely small probability of two customers paying different prices for the same bundle. Furthermore, the presence of the key component in every bundle makes it unlikely that price subadditivity will be violated. This is because the key component is typically a high ticket item such as the base car or base house. Thus, if $B(i) \subseteq B(j) \cup B(k)$, and all three bundles contain the key component, it is very likely that $p_i \leq p_j + p_k$.

The times reported in Table 4 are for step 1 of the algorithm, and are on a Compaq 386 machine with a 387 co-processor with the LINGO modeling language and mixed integer programming software (Cunningham and Schrage 1988). The time required for steps 2–4 was negligible. These times are upper bounds on the actual times used since we were not always able to immediately record when a run was finished.

5. Extensions

In this paper we show how to solve a wide range of bundle pricing problems. Several important areas for model extension exist. One involves costs. For some applications it would be very useful to allow the marginal cost of a bundle to depend on the number and type of other bundles which are offered. A partial solution to this problem is to define setup costs which are incurred if a particular component is present in any bundle. Once this component is produced in any bundle the incremental cost of using it elsewhere is much lower. Making the marginal cost of a bundle dependent on the scope of the entire product line is more difficult, as it introduces nonlinearities into the model.

TABLE 4

21-Component Computational Results

	Problem Characteristics							
	Key Comp					CPU Minutes		
Run	MC	Res Price	Num Cheap	Num Mod	Num Exp	Min	Max	Avg
1–20	65000	80000	10	5	5	4	159	23.55
21–25	65000	80000α	20	0	0	3	8	5.6
26–30	65000	80000α	0	20	0	6	23	11.8
31–35	65000	80000α	0	0	20	2	15	6
36–40	70000	80000α	10	5	5	5	14	8

The assumption of free disposal of unwanted components is used in the proof of subadditivity and single purchases. However, there are circumstances where this assumption is not valid. For example, certain customers strictly prefer an automobile without power windows to one which has power windows. It is not possible for these consumers to dispose of this form of window opening component easily. Determining the implications of a lack of free disposal is important for pricing components in such circumstances. Note, however, that it is quite natural in the model to represent some consumer segments as having negative valuations on certain components.

The essential aspect of the model is maximization of an objective subject to consumer self-selection. As such, our solution techniques have applications other than bundle pricing. Consider the design of product lines with respect to quality. In this circumstance consumers scan the range of products being offered and choose the selection giving them the largest surplus. Interactions between product locations and prices are important, and must be carefully balanced (see Moorthy 1984 and Hanson and Narasimhan 1987). A disjunctive programming formulation might aid in these calculations.

Another example is the design of compensation schemes for both executives and sales personnel. The firm wishes to design its incentives and salaries to maximize profits, while allowing executives or sales personnel to freely choose between compensation schemes (Basu, Lal, Srinivasan, Staelin 1985). This typically results in very difficult computational issues and nonlinearities. Formulating these choices as disjunctive programs, along the lines discussed here, could permit substantial simplification for a range of compensation questions.[1]

[1] The referees offered many useful suggestions which led to the improvement of this manuscript. Zeger Degraeve was instrumental in the computational experimentation. Gary Eppen and Linus Schrage provided many helpful comments.

Dr. Hanson's research was partially funded as the Beatrice Scholar at the University of Chicago Graduate School of Business for 1987–1988.

Dr. Martin's research was partially funded by a grant from the Mathematics Division of the General Motors Research Laboratories.

Appendix

Price Subadditivity Subproblem Notation

$\delta_j = 1$ if the price of bundle j violates subadditivity, 0 otherwise; this is a variable.

$\gamma_{jl} = 1$ if bundle l is one of the bundles assembled in order to obtain the components of bundle j more cheaply than buying bundle j, 0 otherwise; this is a variable.

$E'(L)$ is an index set of bundles formed by taking the union of the bundles in $E(L)$ with all the composite bundles in the solution of (21) to (36).

$e_{jh} = 1$ if component h is in bundle $B(j)$, 0 otherwise; this is a parameter.

$$\max \sum_{j \in E'(L)} p_j \delta_j - \sum_{l \in E'(L)} p_l \gamma_{jl} \tag{37}$$

$$\text{s.t.} \quad e_{jh} \delta_j \leq \sum_{\substack{l \in E'(L) \\ l \neq j}} e_{lh} \gamma_{jl}, \qquad j \in E'(L); h = 1, \ldots, n, \tag{38}$$

$$\delta_j, \gamma_{jl} \in \{0, 1\}. \tag{39}$$

This problem is linear since the prices that appear in the objective function are fixed at their solution value from the solution to problem (21)–(36). If the optimal objective value of (37)–(39) is zero then there is no violation of price subadditivity. If the objective function value is strictly positive then the $\delta_j = 1$ indicate which bundle prices violate subadditivity. Constraint (38) requires the assembled bundle to cover the bundle that violates subadditivity; that is, any component present in a bundle with $\delta_j = 1$ must be present in one of the bundles with $\gamma_{jl} = 1$. If such an assembled bundle is found with a lower price than the price of bundle j subadditivity is violated.

Consumer Surplus Maximization Subproblem Notation

$\pi_{kh} = 1$ if component h is in the bundle which maximizes the surplus of consumer k, 0 otherwise.

$\beta_{kl} = 1$ if consumer k buys bundle l in order to maximize surplus, 0 otherwise.

The optimization problem given below is used to verify that the optimal solution to (21)–(36) maximizes consumer surplus for the given prices.

$$\max \sum_{k=1}^{M} \sum_{h=1}^{n} R_{k\{h\}} \pi_{kh} - \sum_{k=1}^{M} \sum_{l \in E'(L)} p_l \beta_{kl} \tag{40}$$

$$\text{s.t.} \quad \pi_{kh} \leq \sum_{l \in E'(L)} e_{lh} \beta_{kl}, \qquad k = 1, \ldots, M; \qquad h = 1, \ldots, n, \tag{41}$$

$$0 \leq \pi_{kh} \leq 1, \qquad \beta_{kl} \in \{0, 1\}. \tag{42}$$

If the consumer surplus for each consumer k in the solution of (40)–(42) does not exceed the consumer surplus of k in the optimal solution of (21)–(36) then the optimal solution of (21)–(36) maximizes consumer surplus for the given prices. Note that the consumer surplus maximization subproblem is separable in customer segments k, making it easier to optimize. Because this is a maximization problem and reservation prices are nonnegative, integrality of the β_{kl} implies integrality of the π_{kh} variables.

Proofs of Lemmas and Propositions

PROOF OF LEMMA 1. Let the vector of prices $\mathbf{p}^*$ correspond to an optimal solution. Assume that as part of this solution there exist bundles $B(i)$, $B(j)$, $B(l)$ such that $B(i) \subseteq B(j) \cup B(l)$ with $p_i > p_j + p_l$. By free disposal and zero consumer assembly costs, bundle $B(i)$ is not purchased because every consumer receives strictly greater surplus by assembling $B(i)$ from $B(j)$ and $B(l)$. Let $p'_i = p_j + p_l$. Substituting p'_i for p_i cannot affect any other consumer decision because the surplus from consuming the components in $B(i)$ has not changed. Firm profit cannot decrease from substituting p'_i because the marginal costs of production are subadditive. Repeating this argument for any nonsubadditive combination of prices in p^* yields a new subadditive price vector p' with the same profits as p^*. □

PROOF OF PROPOSITION 1. We show that the consumer is never better off by making multiple purchases. Assume the consumer does make more than one purchase and buys bundle $B(i)$ and bundle $B(j)$. Define a new bundle l, by $B(l) = \{h \mid h \in B(i) \cup B(j)\}$. By Lemma 1, the price of $B(l)$ will not exceed the price of $B(i)$ plus the price of $B(j)$. By the assumption of zero marginal utility for any second component the reservation price of bundle $B(l)$ must equal the reservation price of bundle $B(i)$ plus bundle $B(j)$. Therefore the consumer surplus from purchasing $B(l)$ must be at least as great as the consumer surplus from purchasing bundle $B(i)$ and bundle $B(j)$ and the consumer is never better off by making more than one purchase. □

PROOF OF LEMMA 2. It suffices to show there is an optimal solution such that if $B(i) \subset \cup_{j \in I} B(j)$ and $p_i > \sum_{j \in I} p_j$ then no one purchases bundle i. Define bundle l by $B(l) = \{h \mid h \in \cup_{j \in I} B(j)\}$. By free disposal $R_{ki} \leq R_{kl}$ all k. From (12), $p_l \leq \sum_{j \in I} p_j$. Then $R_{ki} - p_i < R_{kl} - \sum_{j \in I} p_j \leq R_{kl} - p_l$ which implies there is an optimal solution where no customer purchases bundle i. □

PROOF OF PROPOSITION 3. Dividing the objective function $\max_i \{g_k(B_i) + \lambda(Y_k - p_i)\}$ by the positive constant λ_k does not change the optimal solution. The income Y_k is a constant and does not affect the optimization.

Thus, maximizing $g_k(B(i))/\lambda_k - p_i$ yields the utility maximum if the utility function has the form given in (17). $\square$

PROOF OF PROPOSITION 4. For sake of brevity, refer to the bundle pricing model using reservation prices R_{ki} as model R and the bundle pricing model using reservation prices Q_{ki} as model Q. First show that the optimal objective function value of model Q is never less than the optimal objective function value of model R. This is done by taking the optimal solution to model R and using it to construct a *feasible* solution to model Q with an equal objective function value.

By condition (ii) it follows that there exists an optimal solution to model R such that if customer segment k selects bundle i then bundle i contains the key component n. Construct a new price vector by increasing the price of every bundle not containing the key component by δ. Any δ such that $s_k \geq Q_{ki} - p_i - \delta$ for all i such that $n \notin B(i)$ is sufficient. Because $Q_{ki} = R_{ki}$ for all bundles i which contain the key component it follows that these revised prices satisfy the consumer surplus constraints in model Q.

Increasing the price of every bundle by a constant except for those bundles which contain the key component maintains price subadditivity. (See constraint set (3) and observe that there is no change in the constraints when $n \in B(i)$. When $n \notin B(i)$ the right-hand side grows by at least 2δ and the left grows by only δ.) Thus the altered prices are feasible in model Q. Since there is no change in revenue or cost data this solution is feasible in both models and yields the same objective function value. Thus the optimal solution value of model Q is at least as great as the optimal solution value of model R.

Now show that an optimal solution to model Q is a feasible solution to model R. Condition (iii) and the fact that $Q_{ki} = R_{ki}$, for all bundles i which contain the key component, imply that $Q_{ki} = R_{ki}$ in all bundles which are selected in the optimal solution to model Q. Condition (i) then implies that this optimal solution satisfies all of the consumer surplus constraints in the model with reservation prices R_{ki}. Therefore, this solution is also optimal to model R. $\square$

PROOF OF PROPOSITION 5. First we show that the relaxed bundle pricing model is indeed a relaxation of the bundle pricing model. To see this observe that any *feasible solution* to the bundle pricing model is feasible to the relaxed bundle pricing model since any bundle can be defined as the composite bundle. Both problems have the same objective function, hence model (21)–(36) is a relaxation of (1)–(11) and part 1 of the proposition is immediate.

If conditions (i) through (iii) of part 2 hold for a feasible solution to the relaxed bundle pricing model then the solution is also feasible to the bundle pricing model. Part 2 then follows by equality of the objective functions. $\square$

References

ADAMS, W. J. AND J. YELLEN, "Commodity Bundling and the Burden of Monopoly," *Quart. J. Economics*, 40 (1976), 475–498.

BALAS, E., "Disjunctive Programming and a Hierarchy of Relaxations for Discrete Optimization Problems," *SIAM J. Algebraic Discrete Methods*, 6 (1985), 466–486.

——— AND R. JEROSLOW, "Cannonical Cuts on the Unit Hypercube," *SIAM J. Appl. Math.*, 31 (1972), 146–164.

BASU, A., R. LAL, V. SRINIVASAN AND R. STAELIN, "Salesforce Compensation Plans: An Agency Theoretic Perspective," *Marketing Sci.*, 4 (1985), 267–291.

BAUMOL, W. J., J. C. PANZER AND R. WILLIG, *Contestable Markets and the Theory of Industry Structure*, Harcourt-Brace-Jovanovich, New York, 1982.

BRATLEY, P., B. L. FOX AND L. E. SCHRAGE, *A Guide to Simulation*, Springer-Verlag, New York, 1983.

CUNNINGHAM, K. AND L. SCHRAGE, "The LINGO Modeling Language," Graduate School of Business, University of Chicago, 1988.

EPPEN, G. D. AND F. J. GOULD, *Quantitative Concepts for Management: Decision Making Without Algorithms*, Prentice Hall, Englewood Cliffs, NJ, 1985.

GOLDBERG, S., P. GREEN AND Y. WIND, "Conjoint Analysis of Price Premiums for Hotel Amenities," *J. Business*, 57 (1984), 111–132.

GREEN, P. AND W. DESARBO, "Componential Segmentation in the Analysis of Consumer Tradeoffs," *J. Marketing*, 43 (1979), 83–91.

——— AND A. KRIEGER, "Models and Heuristics for Product Line Selection," *Marketing Sci.*, 4 (1985), 1–19.

HANSON, W. AND C. NARASIMHAN, "Strategic Quality and Variety Competition," University of Chicago Working Paper, 1987.

HILLIER, F. S. AND G. J. LIEBERMAN, *Introduction to Operations Research*, 4th Ed., Holden-Day, Inc. Oakland, CA, 1986.

JEROSLOW, R. G. AND J. K. LOWE, "Modelling with Integer Variables," *Math. Programming Study*, 22 (1984), 167–184.

——— AND ———, "Experimental Results on the New Techniques for Integer Programming Formulations," *J. Oper. Res. Soc.*, 36 (1985), 393–403.

KALISH, S. AND P. NELSON, "An Empirical Evaluation of Multiattribute Utility and Reservation Price Measurement," Purdue University Working Paper, 1988.

KENNEY, R. AND B. KLEIN, "The Economics of Block Booking," *J. Law and Economics*, 26 (1983), 497–540.

MOORTHY, S. K., "Market Segmentation, Self-Selection and Product Line Design," *Marketing Sci.*, 3 (1984), 288–307.

OGAWA, K., "An Approach to Simultaneous Estimation and Segmentation in Conjoint Analysis," *Marketing Sci.*, 6 (1987), 66–81.

PORTER, M., *Competitive Advantage: Creating and Sustaining Superior Performance*, Free Press, New York, 1985.

SCHMALENSEE, R., "Gaussian Demand and Commodity Bundling," *J. Business*, 57 (1984), S211–S231.

STIGLER, G., "United States v. Loew's, Inc: A Note on Block Booking," *Supreme Court Rev.*, 152 (1963).

TELSER, L. G., "A Theory of Monopoly of Complementary Goods," *J. Business*, 52 (1979), 211–230.

WAGNER, H. M., *Principles of Operations Research*, Prentice-Hall, Englewood Cliffs, NJ, 1975.